

A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Heavy metals: lead, mercury - Set 1

Mark scheme

AQA spec 3.4.3.2.11; Pollution; textbook pp. 275-279

Marking guidance

- The number in brackets after each marking point is the mark it earns; the note above the points says how many are needed.
- Where a question asks for a fixed number of points (eg *two reasons*), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	<i>A correct final answer (with unit where asked) scores full marks. If the answer is wrong, award 1 mark for each correct step of working shown.</i> • $(15.0 - 2.2) / 15.0 \times 100$ (1) • = 85.3% (accept 85%) (1)	2
01.2	<i>1 mark for each point.</i> • Added as an anti-knock agent to smooth combustion and reduce engine wear (1) • Lead particulates were released in exhaust gases into the air of cities and inhaled / deposited on soil and food (1) • Lead is a neurotoxin that bioaccumulates; chronic exposure damages the nervous system / brain development in children (1)	3
01.3	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Replacement of lead water pipes / lead solder with copper / tin-based solder; phosphoric acid dosing of water supplies forming insoluble lead phosphate (1) • Phasing out of lead-based paint / safer paint removal (1) • Lead-free electrical solder (1) • Restrictions on lead fishing weights / shotgun pellets (1) • Industrial emission controls (1)	2
01.4	<i>1 mark for each point.</i> • Reacts with the lead in old pipes to form an insoluble layer of lead phosphate on the inside of the pipe (1) • This prevents lead dissolving into the water (1)	2
01.5	<i>1 mark.</i> • Swans / ducks swallow them (mistaking them for snails / grit) and the lead is absorbed, causing lead poisoning (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 275-276; AQA spec 3.4.3.2.11.	

Question	Marking guidance	Marks
02.1	<i>1 mark for each point.</i> • Inorganic mercury settled into anaerobic sediments where anaerobic microbes converted it to organic (methyl) mercury (1) • Methyl mercury is liposoluble so it is absorbed easily and stored in fat / passes through cell membranes (1) • It bioaccumulated in fish and shellfish and biomagnified along the food chain to reach much higher concentrations than in the water (1) • Local people ate large amounts of fish (critical group) so received toxic doses (1)	4
02.2	<i>1 mark for each point.</i> • Liposoluble so it crosses the blood-brain barrier into the brain, inhibiting enzymes / nerve function (neurotoxin) (1) • It crosses the placenta and harms the developing unborn baby (1) • Mercury is a teratogen, preventing normal gene expression / development (1)	3
02.3	<i>1 mark for each point.</i> • Elemental (liquid) mercury: not easily absorbed through skin or gut, though vapour is absorbed in the lungs (1) • Inorganic compounds (eg mercury oxide): moderately well absorbed in the gut (1) • Organic compounds (eg methyl mercury): easily absorbed through skin, gut and lungs; liposoluble so most toxic and bioaccumulates (1)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 276-277; AQA spec 3.4.3.2.11.	
03.1	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Disposal of items containing mercury: batteries, thermometers, fluorescent lights (1) • Chemical plants producing chlorine using mercury electrodes (1) • Combustion of coal (1) • Gold mining (amalgamation) (1)	3
03.2	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Replacement of mercury thermometers with alcohol / electronic thermometers (discontinued uses) (1) • Removal from effluents by reverse osmosis / activated carbon filters / ion exchange filters (1) • Disposal / storage of wastes at high pH to reduce solubility (1) • Encapsulation of mercury-containing hazardous waste in cement (1)	3
03.3	<i>1 mark for each point.</i> • Most heavy metals are more soluble at low pH / less soluble at high pH (1) • So at high pH they remain insoluble / immobile and cannot be leached into water or absorbed by organisms (1)	2
03.4	<i>1 mark for each point.</i> • The effects of two pollutants interact to produce a different (greater) effect than either alone (1) • Cadmium and zinc; ozone and sulfur dioxide (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 275-278; AQA spec 3.4.3.2.11.	
04.1	<i>1 mark for the correct answer.</i> • $180 / 1.2 = 150$ (1)	1

Question	Marking guidance	Marks
04.2	<i>1 mark.</i> • Cadmium and zinc occur together in the same ores so mine drainage / smelting releases both (1)	1
04.3	<i>1 mark for each point.</i> • Mine drainage water contaminated irrigation water / paddy soil so rice absorbed cadmium (1.2 vs 0.05 mg kg^{-1}) (1) • Cadmium is liposoluble / not excreted so it bioaccumulates in the body (kidneys) with chronic exposure to small doses over years (1) • Cadmium causes kidney failure and bone decalcification / skeletal collapse (itai-itai disease) (1) • Rice is the staple food so residents receive a large dose; village B has no exposure and no disease (correlation supports causation) (1)	4
04.4	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Collect mine drainage water and neutralise / raise pH with lime so cadmium precipitates and stays insoluble (1) • Cover / cap spoil heaps to prevent leaching and dust (1) • Recycle nickel-cadmium batteries / hazardous waste landfill (1) • Phytoremediation of contaminated paddy soils (1)	2
04.5	<i>1 mark for each point.</i> • Manufacture may release cadmium in effluents / dust (1) • Panels have a limited life; disposal of old panels (landfill, incineration of pigments) could release cadmium unless recycled (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 275-279; AQA spec 3.4.3.2.11.	
05.1	<i>Correct answer only.</i> • B (1)	1
05.2	<i>1 mark for each point.</i> • TBT is highly toxic to marine molluscs / crustaceans and an endocrine disruptor altering growth and reproduction of oysters / whelks even at very low concentrations (1) • It bioaccumulated in shellfish and could pass to humans eating them (1) • It was assumed dilution would make it harmless (illustrates the need for the precautionary principle); now prohibited and replaced by copper (1)	3
05.3	<i>1 mark for each point.</i> • In the low-oxygen conditions of the spoil heap iron is in its soluble reduced form and is leached into the river (1) • In the river it is oxidised to insoluble iron oxide (orange sediment); the oxidation uses dissolved oxygen so the river is deoxygenated, killing aerobic organisms (1) • Control: drainage water is passed over mesh screens where the iron oxidises and is deposited before the water reaches the river; the solid iron is removed periodically (1)	3
05.4	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Lead-acid car batteries (recycled) (1) • Lead flashing on roofs / windows (1) • Radiation shielding (1)	2
05.5	<i>1 mark.</i> • They inhibit enzyme function, especially the enzymes of the nervous system / nerve cells (1)	1

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 275-279; AQA spec 3.4.3.2.11.	
06.1	<i>Correct answer only.</i> • Organic (methyl) mercury (1)	1
06.2	<i>1 mark for each point.</i> • Anaerobic microbes converted the inorganic mercury into organic methyl mercury (1) • Methyl mercury is liposoluble so it is absorbed, crosses into the brain and biomagnifies along the food chain to fish eaten by people (1)	2
06.3	<i>1 mark for each point.</i> • Most heavy metals are less soluble at high pH (1) • So they remain immobile and are not leached into water or absorbed by organisms (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 275-279; AQA spec 3.4.3.2.11.	